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saxsabs: Absolute-intensity calibration and provenance tracking for small-angle X-ray scattering

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Software

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Summary

Small-angle X-ray scattering (SAXS) profiles are easier to compare across measurements when their intensities are reported on an absolute scale. saxsabs estimates the calibration factor K from a measured reference, such as NIST Standard Reference Material (SRM) 3600 glassy carbon or water at a documented temperature (Allen et al., 2017; National Institute of Standards and Technology, 2016; Orthaber et al., 2000). It normalizes intensities using exposure, monitor, and transmission measurements, records sample thickness and prior corrections, and exports calibrated profiles as text, canSAS1d XML, or NXcanSAS HDF5 (canSAS Working Group, 2013; NeXus International Advisory Committee, 2026).

The Python API and command line support reusable calculations and external one-dimensional (1D) profiles. The bilingual Workbench provides interactive calibration and batch operations, while a separate strict workflow processes two-dimensional detector data under SPring-8 BL19B2 conventions. The software stops scaling or subtraction when required physical inputs or the profile's intensity state cannot be established.

Statement of need

Absolute SAXS intensities, commonly reported in cm^{-1}, support quantitative comparisons between samples and experiments. Their scale depends on monitor and transmission normalization, sample thickness, and calibration against a known reference (Allen et al., 2017; Orthaber et al., 2000). In practice, reduced profiles often arrive from different programs with inconsistent metadata and processing histories. If a profile's current scale is unclear, applying a calibration or thickness correction twice can produce a plausible curve on the wrong intensity scale.

saxsabs serves beamline scientists and SAXS users who need to calibrate external 1D profiles or process detector data while recording the inputs and corrections behind each result. It distinguishes raw counts, relative intensity, absolute intensity in cm^{-1}, and ambiguous states. Calibration and subtraction proceed only when the declared state and required metadata agree. The strict 2D workflow handles current BL19B2 data conventions; reusable calculation and I/O modules also support other interfaces.

State of the field

Existing tools cover important neighboring tasks. pyFAI performs azimuthal integration of detector images (Ashiotis et al., 2015), and FabIO reads two-dimensional detector formats (Knudsen et al., 2013). SasView and Irena support small-angle scattering analysis and model fitting (Ilavsky & Jemian, 2009; Shrestha et al., 2026). BioXTAS RAW combines BioSAXS

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reduction with water- or glassy-carbon-based absolute scaling and buffer subtraction (Hopkins et al., 2017). These packages remain appropriate for the tasks they were designed to solve.

The materials-scattering work that prompted `saxsabs` required a programmatic way to calibrate 1D profiles from different reduction programs and a strict BL19B2 path that checks correction history before campaign scaling. A separate package makes it possible to apply those checks without changing upstream reduction or fitting tools: BL19B2 metadata rules stay within the campaign workflow, while detector integration and image reading remain delegated to pyFAI and FabIO.

Software design

The package separates scientific calculations and file I/O from its user interfaces (Figure 1). The Python API and CLI expose reusable normalization, calibration, parsing, subtraction, and export functions. The Workbench provides interactive calibration and batch operations. The strict BL19B2 runner owns campaign processing; these interfaces share calculations and I/O but do not claim identical workflows.

Before an absolute-scale operation, `saxsabs` assesses the profile's state from its metadata, units, column names, and correction ledger. It distinguishes `raw_counts`, `relative`, `absolute_cm^-1`, and `ambiguous`. Conflicting or missing evidence remains ambiguous, so the software refuses operations that could repeat thickness or K corrections. The `corrections_applied` ledger records completed operations; a separate `do_not_repeat` field acts only as an execution guard and cannot establish that a correction was physically applied.

For calibration, the measured standard curve is interpolated onto the reference grid, and pointwise reference-to-measured intensity ratios are calculated. The estimator takes the median ratio after rejecting outliers by the median absolute deviation. For the built-in SRM 3600 reference, a separate curve-parallelism quality-control step uses the certificate's maximum expanded relative-intensity uncertainty (6.25%) as the default tolerance before outlier filtering. This is a project-defined heuristic, not a NIST per-point acceptance limit; users can set a stricter tolerance. The synthetic example in Figure 2 illustrates the robust filtering step using a fixed-seed profile and explicitly supplied SRM 3600 reference data; it does not represent the built-in certificate-check path. The ratio-scatter term is an approximate standard error of the median, assuming independent, normally distributed inlier ratios; it does not model measurement noise or cross-q covariance. When reference uncertainty is available, it is combined in quadrature with this term and reported as a partial K-factor uncertainty. In the BL19B2 workflow, the software reports a partial combined uncertainty when shared covariance is unquantified and does not report a system expanded uncertainty.

The strict BL19B2 workflow validates detector, monitor, transmission, thickness or attenuation, reference, and output inputs before reduction and calibration. It keeps required beamline metadata explicit. For attenuation, the workflow uses a fixed 30 keV NIST SRD 126 table, while a separate diagnostic calculator uses energy-dependent Elam data through xraydb (Elam et al., 2002; Hubbell & Seltzer, 2004; Newville et al., 2025). Writers support plain text and the canSAS1d 1.1 and NXcanSAS layouts (canSAS Working Group, 2013; NeXus International Advisory Committee, 2026). The deterministic 9×9 synthetic detector example checks reduction through export and recovers its planted K and sample curve; automated tests cover core calculations, parsers, exporters, CLI behavior, and Workbench validation rules.

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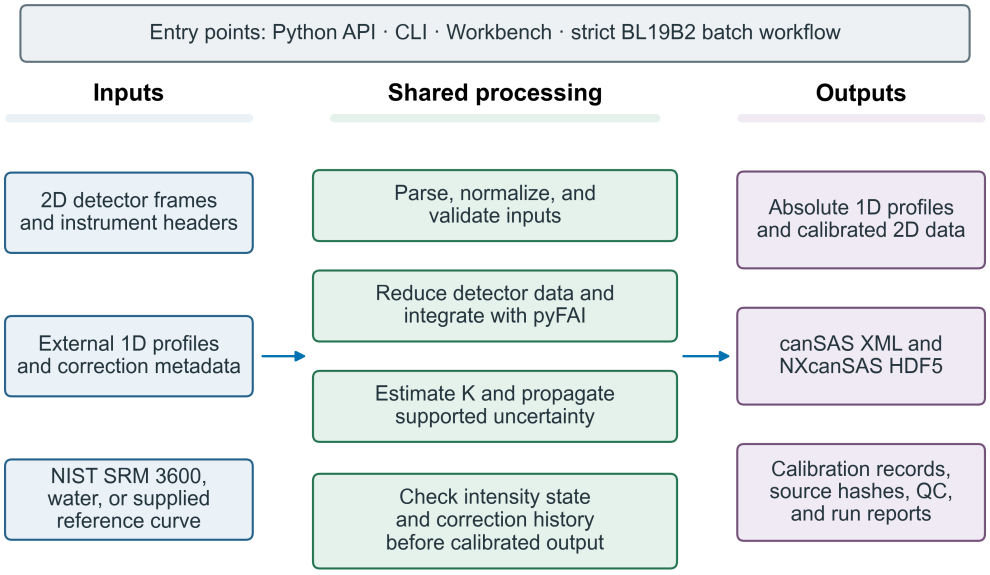


Figure 1: Package architecture and data flow. The Python API, command line, Workbench, and strict BL19B2 workflow use the scientific and I/O modules to validate inputs, calibrate profiles, and write absolute-scale outputs with processing records.

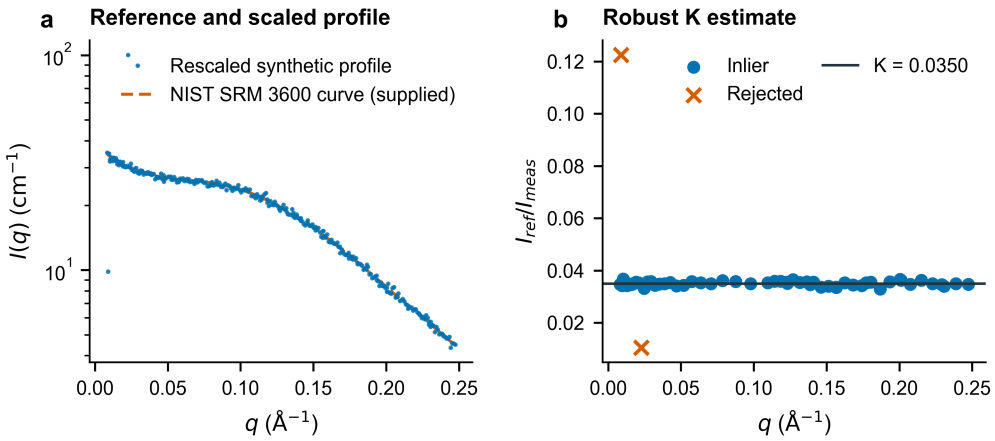


Figure 2: Robust K estimation with deterministic synthetic data. Panel (a) shows the NIST SRM 3600 certificate curve and a fixed-seed synthetic measured profile rescaled by the planted K. Panel (b) shows the reference-to-measured ratios and two injected outliers rejected by median/MAD filtering. The certificate values are passed as an explicit user-supplied reference.

Software availability

The source code, tests, documentation, and examples are available in the AbsSAXS GitHub repository under the BSD-3-Clause license. The `main` branch contains the unreleased 2.0.0 candidate submitted for review; GitHub Release v1.1.1 is an earlier version. Earlier archived releases are indexed by the Zenodo concept record (Gong, 2026).

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Research impact statement

At SPring-8 BL19B2, the author has used saxsabs to prepare absolute-scale SAXS/USAXS profiles for the published spinodally modulated Ti-24Nb-4Zr-8Sn study (Gong et al., 2026) and for ongoing metallic-materials work. The article does not cite saxsabs; the processing records are available to the editor on request.

AI usage disclosure

Earlier project work used GitHub Copilot, Anthropic Claude, OpenAI Codex, and xAI Grok only for coding assistance, including code refactoring and test scaffolding; exact versions of those earlier tools were not retained. During this submission-preparation pass, OpenAI Codex (GPT-6) assisted with manuscript editing, figure scripts, input validation and tests, packaging and CI updates, README and documentation changes, and reference verification. OpenAI image generation, with a version not exposed by the service, produced the conceptual README cover. That cover is illustrative and contains no experimental data. Figures in this paper are rendered by project scripts from the stated reference data and synthetic inputs; the Workbench screenshot is a direct application capture. Software descriptions were checked against the implementation and project documentation; targeted tests were run; and reference metadata and DOIs were checked against publisher, NIST, and Zenodo records. Example inputs and scripts were checked to distinguish synthetic demonstrations from experimental validation. The author reviewed, edited, and validated all AI-assisted code, documentation, manuscript text, and images; made the core scientific and software design decisions; and accepts full responsibility for the software and manuscript.

Author contributions

Delun Gong: Conceptualization, Data curation, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing - original draft, and Writing - review and editing.

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